A Framework of Greater India

Cyclotomic Economic Field Theory with 3720 Districts

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Abstract

This treatise develops a mathematical framework for studying cooperation among thirteen Asian nations through cyclotomic polynomials, Euler's totient function, and an aggregate polynomial construction. It connects the resulting mathematical structure to economic coordination, statistical harmonisation, financial planning, institutional design, international relations, and interdisciplinary research.

The construction yields thirteen selected cyclotomic indices whose sum is 3720. The product of the corresponding cyclotomic polynomials has degree 2976. The distinction between the sum of indices and the degree of the product is fundamental: neither quantity, by itself, establishes the number of administrative districts or the feasibility of a particular political arrangement.

The proposed framework is mathematical and exploratory. Its institutional applications require independent legal analysis, empirical validation, democratic legitimacy, reciprocity, and respect for the sovereignty of participating states.

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1 Purpose and Scope

The phrase *Greater India* is used here as the name of a proposed framework for voluntary regional cooperation, not as a claim of territorial ownership or political authority over any present-day state.

The thirteen countries represented in the mathematical exercise are India, Indonesia, Afghanistan, China, Thailand, Pakistan, Malaysia, Myanmar, Nepal, Bangladesh, Sri Lanka, Maldives, and Bhutan.

The central methodological principle is that mathematical coordination does not require institutional uniformity. Countries may cooperate through common statistical standards, compatible economic models, negotiated rules, and reciprocal arrangements while retaining their own constitutional systems, currencies, administrative boundaries, and policy preferences.

The construction has four principal numerical objects:

$$N_{\text{countries}} = 13, \tag{1}$$

$$\sum_{j=1}^{13} n_j = 3720, \tag{2}$$

$$\sum_{j=1}^{13} \varphi(n_j) = 2976, \tag{3}$$

$$\deg\left(\prod_{j=1}^{13} \Phi_{n_j}(x)\right) = 2976. \tag{4}$$

Here n_j is a selected cyclotomic index, $\Phi_{n_j}(x)$ is the corresponding cyclotomic polynomial, and φ is Euler's totient function.

The subtitle's number 3720 denotes the sum of the selected indices. It is not the sum of the adjusted polynomial degrees. The latter is 2976. The title is retained as requested, while the mathematical interpretation is made explicit.

2 Mathematical Foundations

2.1 Cyclotomic polynomials

The n-th cyclotomic polynomial is defined by

$$\Phi_n(x) = \prod_{\substack{1\le k\le n\\ \gcd(k,n)=1}} \left(x - e^{2\pi i k/n}\right). \tag{5}$$

Its roots are precisely the primitive n-th roots of unity. A fundamental identity is

$$x^n - 1 = \prod_{d\mid n} \Phi_d(x). \tag{6}$$

Cyclotomic polynomials are monic and belong to $\mathbb{Z}[x]$. Their degrees satisfy

$$\deg \Phi_n(x) = \varphi(n), \tag{7}$$

where Euler's totient function counts the integers between 1 and n that are relatively prime to n [1, 2].

For the prime factorisation

$$n = \prod_{i=1}^{r} p_i^{a_i}, \tag{8}$$

Euler's totient formula is

$$\varphi(n) = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right). \tag{9}$$

For a prime p,

$$\Phi_p(x) = 1 + x + \cdots + x^{p-1}. \tag{10}$$

For distinct primes p and q,

$$\Phi_{pq}(x) = \frac{(x^{pq}-1)(x-1)}{(x^p-1)(x^q-1)}. \tag{11}$$

These identities allow the relevant polynomial degrees to be verified without expanding large polynomials into individual coefficients.

2.2 The degree-selection problem

For a desired degree D, the selection problem is to find the least positive integer n satisfying

$$\varphi(n) = D. \tag{12}$$

Such an n need not exist for every positive integer D. In particular, $\varphi(n)$ is even for every $n > 2$. Consequently, no cyclotomic polynomial has degree 401.

The original target degrees for Indonesia and Afghanistan are therefore adjusted downward by one at a time until a totient value is found. This procedure gives the adjusted degrees 512 and 400, respectively.

3 The Thirteen-Nation Construction

3.1 Selected indices and polynomial degrees

The following table records the selected indices and degrees. The degree values for Indonesia and Afghanistan reflect the specified decrement rule.

Table 1: Selected indices and polynomial degrees.

No.	Country	Index n_j	Degree $\varphi(n_j)$
1	India	1025	800
2	Indonesia	771	512
3	Afghanistan	401	400
4	China	501	332
5	Thailand	283	282
6	Pakistan	173	172
7	Malaysia	187	160
8	Myanmar	143	120
9	Nepal	73	72
10	Bangladesh	85	64
11	Sri Lanka	35	24
12	Maldives	25	20
13	Bhutan	19	18
	Total	**3720**	**2976**

For each country, the polynomial is $\Phi_{n_j}(x)$. The following identities verify the selected degrees:

$$\varphi(1025) = \varphi(5^2 \cdot 41) = 800, \tag{13}$$
$$\varphi(771) = \varphi(3 \cdot 257) = 2 \cdot 256 = 512, \tag{14}$$
$$\varphi(401) = 400, \tag{15}$$
$$\varphi(501) = \varphi(3 \cdot 167) = 2 \cdot 166 = 332, \tag{16}$$
$$\varphi(283) = 282, \tag{17}$$
$$\varphi(173) = 172, \tag{18}$$
$$\varphi(187) = \varphi(11 \cdot 17) = 10 \cdot 16 = 160, \tag{19}$$
$$\varphi(143) = \varphi(11 \cdot 13) = 10 \cdot 12 = 120, \tag{20}$$
$$\varphi(73) = 72, \tag{21}$$
$$\varphi(85) = \varphi(5 \cdot 17) = 4 \cdot 16 = 64, \tag{22}$$
$$\varphi(35) = \varphi(5 \cdot 7) = 4 \cdot 6 = 24, \tag{23}$$
$$\varphi(25) = 25 - 5 = 20, \tag{24}$$
$$\varphi(19) = 18. \tag{25}$$

The primality of the relevant prime indices and the standard multiplicativity of φ for coprime arguments establish these identities.

3.2 Explicit polynomial representations

For compactness, define the geometric sum

$$S_m(x) = 1 + x + \cdots + x^{m-1} = \frac{x^m - 1}{x - 1}. \tag{26}$$

The thirteen polynomials can be represented as follows:

$$\Phi_{1025}(x) = \frac{S_{41}(x^{25})}{S_{41}(x^5)}, \tag{27}$$

$$\Phi_{771}(x) = \frac{(x^{771}-1)(x-1)}{(x^{257}-1)(x^3-1)}, \tag{28}$$

$$\Phi_{401}(x) = S_{401}(x), \tag{29}$$

$$\Phi_{501}(x) = \frac{S_{167}(x^3)}{S_{167}(x)}, \tag{30}$$

$$\Phi_{283}(x) = S_{283}(x), \tag{31}$$

$$\Phi_{173}(x) = S_{173}(x), \tag{32}$$

$$\Phi_{187}(x) = \frac{S_{17}(x^{11})}{S_{17}(x)}, \tag{33}$$

$$\Phi_{143}(x) = \frac{S_{13}(x^{11})}{S_{13}(x)}, \tag{34}$$

$$\Phi_{73}(x) = S_{73}(x), \tag{35}$$

$$\Phi_{85}(x) = \frac{S_{17}(x^5)}{S_{17}(x)}, \tag{36}$$

$$\Phi_{35}(x) = \frac{S_7(x^5)}{S_7(x)}, \tag{37}$$

$$\Phi_{25}(x) = x^{20} + x^{15} + x^{10} + x^5 + 1, \tag{38}$$

$$\Phi_{19}(x) = S_{19}(x). \tag{39}$$

Each rational expression above simplifies to a polynomial in $\mathbb{Z}[x]$. The expressions are compact representations, not claims that the resulting polynomials have fractional coefficients.

3.3 The aggregate polynomial

Define

$$\begin{aligned}\mathcal{P}(x) = {} & \Phi_{1025}(x)\Phi_{771}(x)\Phi_{401}(x)\Phi_{501}(x)\Phi_{283}(x) \\ & \cdot \Phi_{173}(x)\Phi_{187}(x)\Phi_{143}(x)\Phi_{73}(x)\Phi_{85}(x) \\ & \cdot \Phi_{35}(x)\Phi_{25}(x)\Phi_{19}(x).\end{aligned} \tag{40}$$

Since each factor is monic and belongs to $\mathbb{Z}[x]$, their product also belongs to $\mathbb{Z}[x]$ and is monic. The degree of a product of nonzero polynomials is the sum of their degrees. Hence,

$$\deg \mathcal{P}(x) = \sum_{j=1}^{13} \deg \Phi_{n_j}(x) \tag{41}$$

$$= \sum_{j=1}^{13} \varphi(n_j) \tag{42}$$

$$= 2976. \tag{43}$$

Therefore,

$$\boxed{\mathcal{P}(x) \in \mathbb{Z}[x], \qquad \mathcal{P}(x) \text{ is monic}, \qquad \deg \mathcal{P}(x) = 2976.} \tag{44}$$

4 The Index Sum and the Difference

The sum of the thirteen indices is

$$\begin{aligned}\sum_{j=1}^{13} n_j = {} & 1025 + 771 + 401 + 501 + 283 + 173 \\ & + 187 + 143 + 73 + 85 + 35 + 25 + 19 \end{aligned} \tag{45}$$

$$= 3720. \tag{46}$$

Subtracting the aggregate degree gives

$$3720 - 2976 = \boxed{744}. \tag{47}$$

This difference is a derived integer. It is not automatically the degree of a polynomial or a count of administrative units. It becomes mathematically significant in the present construction because it provides an index for another cyclotomic polynomial.

5 The Cyclotomic Polynomial $\Phi_{744}(x)$

5.1 Prime factorisation and degree

The prime factorisation is

$$744 = 2^3 \cdot 3 \cdot 31. \tag{48}$$

Euler's totient formula yields

$$\varphi(744) = 744\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{31}\right) \tag{49}$$

$$= 744 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{30}{31} \tag{50}$$

$$= \boxed{240}. \tag{51}$$

Consequently,

$$\deg \Phi_{744}(x) = 240, \tag{52}$$

not 744.

5.2 A compact identity

For odd $m > 1$ and integer $a \geq 2$, the relevant cyclotomic identity is

$$\Phi_{2^a m}(x) = \Phi_{2m}\left(x^{2^{a-1}}\right). \tag{53}$$

Since $744 = 2^3 \cdot 93$, this gives

$$\boxed{\Phi_{744}(x) = \Phi_{186}(x^4)}. \tag{54}$$

The degree is consistent because

$$\deg \Phi_{186}(x^4) = 4\varphi(186) = 4 \cdot 60 = 240. \tag{55}$$

5.3 Correct rational factorisation

The general Möbius formula is

$$\Phi_n(x) = \prod_{d \mid n} (x^d - 1)^{\mu(n/d)}, \tag{56}$$

where μ is the Möbius function. Applying it to $744 = 2^3 \cdot 3 \cdot 31$ gives

$$\boxed{\Phi_{744}(x) = \frac{(x^{744} - 1)(x^{124} - 1)(x^{12} - 1)(x^8 - 1)}{(x^{372} - 1)(x^{248} - 1)(x^{24} - 1)(x^4 - 1)}.} \tag{57}$$

Although written as a quotient, this expression simplifies exactly to the monic polynomial $\Phi_{744}(x)$ in $\mathbb{Z}[x]$.

The degree 240 is compatible with the index difference 744, but the two numbers have different mathematical meanings.

6 A Computational Verification Strategy

A reproducible computational procedure can search for the least index associated with a given attainable degree. For finite verification, Euler's totient values may be precomputed using a sieve. The following pseudocode specifies the search logic.

```
INPUT:
    target degree D
    maximum index M

COMPUTE phi[n] for n = 1,...,M using a totient sieve

FOR n = 1,...,M:
    IF phi[n] = D:
        RETURN n

RETURN "No solution found within the search bound"
```

For a decrementing target, the outer procedure is:

```
INPUT:
    original degree D
    maximum index M

FOR d = D, D-1, D-2, ... , 1:
    n = least index with phi[n] = d
    IF such n exists within the verified range:
        RETURN (d, n)

RETURN "No solution found within the search bound"
```

A bounded search does not, by itself, prove that no solution exists above the bound. Claims of global minimality require either a sufficiently rigorous exhaustive argument or a mathematical proof that rules out every smaller index.

For the two original exceptional targets, stronger elementary arguments are available.

First, 401 cannot be a totient value because it is odd and greater than 2. The next value, 400, is attained by the prime index 401:

$$\varphi(401) = 400. \tag{58}$$

Second, $514 = 2 \cdot 257$ is not a totient value. If $\varphi(n) = 514$, every odd prime divisor q of n must satisfy $q - 1 \mid 514$. The only possible values of $q - 1$ are 2 and 514, so q would have to be 3 or 515. Since 515 is composite, the only possible odd prime divisor is 3. Such an n cannot have a totient divisible by 257, a contradiction.

The index $771 = 3 \cdot 257$ attains the decremented degree:

$$\varphi(771) = \varphi(3)\varphi(257) = 512. \tag{59}$$

These arguments establish the decrement outcomes. The least-index claims for the other target degrees should be accompanied by the stated search procedure or independent proofs of minimality in any formal computational appendix.

7 A Conceptual Architecture for Regional Cooperation

The polynomial construction is a mathematical representation. It does not independently determine national policy, institutional authority, or economic outcomes. Its proposed use is to provide a common formal object around which distinct models and measurements can be organised.

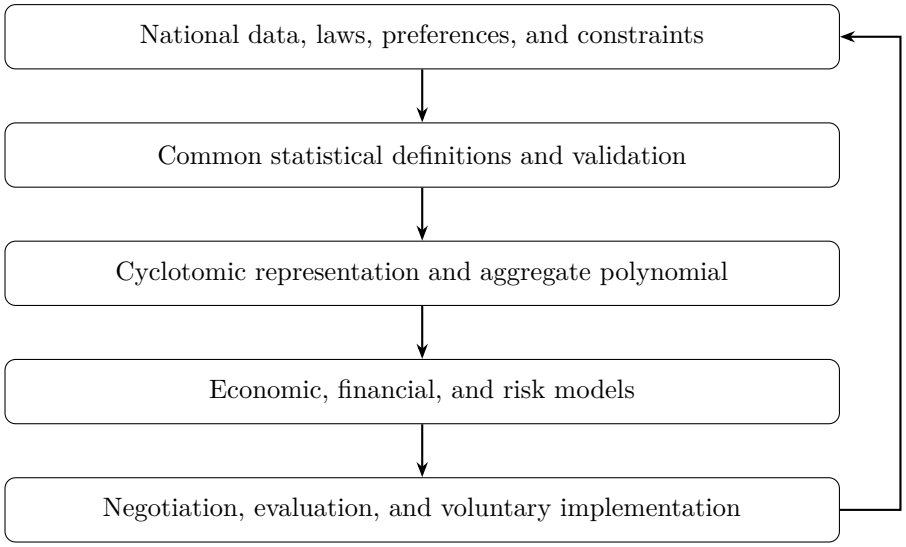


Figure 1: A Conceptual Architecture for Regional Cooperation.

The feedback loop is essential. New data, policy outcomes, legal constraints, and observed risks must be able to revise the modelling assumptions.

8 Economics and Welfare Analysis

8.1 Economic coordination

A regional economic framework can use common definitions for output, productivity, trade, employment, inflation, public investment, and distributional outcomes. Statistical comparability does not require identical national economic structures.

A stylised regional objective function may be written as

$$\max_{\mathbf{a}} \; W(\mathbf{a}) = \sum_{i=1}^{13} \omega_i U_i(\mathbf{a}) - \lambda R(\mathbf{a}), \tag{60}$$

where $\mathbf{a}$ denotes the set of agreed policy actions, U_i represents country-specific welfare, ω_i is an explicitly negotiated welfare weight, and R represents systemic risk. The parameters must not be selected arbitrarily or treated as scientifically established merely because the objective function is written mathematically.

Welfare analysis should examine both aggregate gains and their distribution. A positive sum of gains does not imply that every country, region, industry, or household benefits. Compensation, transitional assistance, public goods, and adjustment costs must be analysed explicitly [3, 4].

8.2 Financial planning and analysis

A regional financial model could compare public investment, infrastructure financing, debt sustainability, trade settlement, contingent liabilities, and alternative allocation rules.

For each proposed project, a basic discounted cash-flow measure is

$$\mathrm{NPV} = \sum_{t=0}^{T} \frac{\mathbb{E}[CF_t]}{(1+r)^t}, \tag{61}$$

where CF_t is the relevant net cash flow and r is an appropriate discount rate. For public projects, the analysis should also report externalities, distributional effects, uncertainty, and the difference between financial and social returns.

The polynomial degree is not a financial return, a budget, a capital requirement, or a measure of national wealth. Any mapping from the mathematical construction to a financial allocation rule requires separate assumptions, units, data, and validation.

9 Econometrics, Statistics, and Data Science

A common regional data architecture could harmonise variable definitions, units, reporting periods, missing-data conventions, and quality-control procedures.

A general panel-data model might take the form

$$y_{it} = \alpha_i + \delta_t + \boldsymbol{\beta}^{\mathsf{T}} \mathbf{x}_{it} + \varepsilon_{it}, \tag{62}$$

where i indexes a country or region, t indexes time, α_i captures unit-specific effects, and δ_t captures common time effects.

Such a model requires assumptions about sampling, measurement, dependence, identification, and error structure. A cyclotomic polynomial does not establish causal identification or guarantee statistical efficiency.

Data-science workflows should include provenance, reproducibility, version control, missingness analysis, sensitivity testing, and independent validation. Comparisons should disclose uncertainty instead of presenting point estimates as exact facts.

10 Algorithms, Computer Science, and AI/ML

The totient-search procedure is an example of a discrete number-theoretic algorithm. Its correctness depends on the implementation of the totient sieve, the search order, and the scope of the claimed result.

For an implementation intended for public policy, useful engineering requirements include:

- deterministic, reproducible calculations;
- unit tests for the totient function and polynomial degrees;
- independent verification of the reported index values;
- explicit search bounds and documented assumptions;
- versioned source code and machine-readable output;
- review of numerical and symbolic computations.

Machine-learning systems may assist with forecasting, anomaly detection, scenario analysis, and data validation. They should not be presumed unbiased or used as substitutes for political consent, legal authority, or accountable human decision-making. Model drift, data leakage, distribution shift, and unequal error rates should be monitored.

11 Physics, Chemistry, and Biology

Cyclotomic polynomials arise naturally in the mathematics of roots of unity and finite periodic structures. Related algebraic ideas appear in discrete Fourier analysis, signal processing, symmetry, and the study of periodic systems.

In chemistry, symmetry and group-theoretic methods can help describe molecular structures and vibrational modes. In biology, mathematical models of periodic processes may involve harmonic analysis, cyclic dynamics, and statistical time-series methods.

These are connections between mathematical methods, not evidence that the present thirteen-polynomial construction directly models physical, chemical, or biological systems. A substantive application would require a domain-specific mechanism, suitable data, testable predictions, and independent empirical support.

12 Linguistics, Psychology, and Psychiatry

Regional coordination may require multilingual public information, translation standards, terminology management, and accessible communication. Linguistic diversity should be treated as an operational and civic consideration rather than a defect to be eliminated.

Psychological research can inform deliberation, trust, cooperation, risk perception, and responses to uncertainty. Psychiatry is a clinical discipline and should not be used as a metaphor for political disagreement or as a means of pathologising dissent.

Participation and consultation should protect privacy, informed consent, and the right to disagree. The mathematical construction provides no basis for diagnosing individuals, populations, or political movements.

13 Law, Political Science, and Diplomacy

Any regional arrangement must be assessed under the relevant domestic constitutional systems, applicable international obligations, and the terms of any treaties actually concluded. The mathematical construction confers no sovereignty, territorial title, legislative competence, or treaty-making authority.

The Charter of the United Nations provides an important general reference for sovereign equality, peaceful relations, and the prohibition on the threat or use of force contrary to the Charter [5]. The precise legal consequences of a proposed institution depend on its design, the parties, and the applicable law.

A defensible diplomatic framework should include:

- voluntary participation and an agreed procedure for withdrawal;
- sovereign equality and reciprocal obligations;
- peaceful dispute settlement;
- transparent decision-making and independent review;
- protection of minorities and affected communities;
- clear limits on delegated institutional powers;
- safeguards against coercion and unilateral domination.

Historical and civilisational connections may support cultural exchange, scholarship, trade, and mutual understanding. They do not independently establish present-day territorial claims or the consent of contemporary populations.

14 Geopolitics, International Relations, and Security

Regional cooperation can reduce some coordination costs while creating new dependencies. The appropriate analysis should therefore consider both benefits and vulnerabilities.

Potential areas of cooperation include disaster response, public health, infrastructure resilience, trade facilitation, scientific exchange, environmental protection, and peaceful conflict prevention. Each area requires its own legal mandate, funding arrangement, operational plan, and evaluation criteria.

Warfare economics can contribute to analysis of the opportunity cost of conflict, defence expenditure, logistics, disruption of trade, civilian harm, and post-conflict reconstruction. It should not be used to treat military escalation as an inevitable consequence of regional integration. Security planning should prioritise de-escalation, civilian protection, accountability, and lawful conduct.

No strategic or military conclusion follows from the sum of cyclotomic indices. Such conclusions require independent geopolitical evidence and scenario analysis.

15 Cybersecurity and Institutional Resilience

A shared data infrastructure would create responsibilities for access control, privacy, integrity, availability, and incident response. A practical security architecture should use least-privilege access, authenticated identities, encryption where appropriate, secure backups, audit logs, vulnerability management, and documented recovery procedures.

A common mathematical representation is not a reason to create a single point of failure. Where appropriate, systems should retain national control over sensitive data and define carefully the conditions under which information may be shared.

Cybersecurity assessment should include threat modelling, independent testing, supply-chain review, and plans for service continuity. Cross-border data exchange must also comply with applicable legal and regulatory requirements.

16 Philosophy, Religion, and Ethical Foundations

The mathematical exercise can be understood as an illustration of coordination without requiring every participant to have the same internal structure. Each country retains its own policies, institutions, traditions, and constraints while contributing to a common formal construction.

Philosophical analysis should distinguish mathematical validity, empirical truth, legal legitimacy, and moral justification. These are related but separate standards of evaluation.

Religious and philosophical traditions may provide ethical resources for reflection on reciprocity, restraint, responsibility, peace, and the welfare of others. No single religious or philosophical interpretation should be presumed to bind all participants. A pluralistic framework should protect freedom of conscience and equal civic standing.

17 Limits, Risks, and Validation

The following limitations are central to any responsible application of the construction.

1. **Mathematical limitation.** The index sum and polynomial degree are different quantities. The value 744 is their difference, while $\deg \Phi_{744} = 240$.

2. **Interpretive limitation.** A polynomial degree does not, by itself, represent a population, territory, budget, or number of administrative districts.

3. **Empirical limitation.** Economic, financial, and social applications require observable variables, credible identification strategies, uncertainty estimates, and out-of-sample validation.

4. **Institutional limitation.** A mathematical framework cannot create political consent, treaty obligations, or lawful authority.

5. **Distributional limitation.** Aggregate welfare improvements may conceal losses to particular groups. Distributional analysis and safeguards are necessary.

6. **Computational limitation.** Search-based minimality claims must specify their bounds and verification methods. A finite search should not be described as a universal proof without justification.

A sound validation programme should separate symbolic verification, computational replication, empirical testing, legal review, and democratic consultation. Success in one category does not automatically establish success in the others.

18 Conclusion

The thirteen selected cyclotomic indices sum to 3720, and the product of the corresponding thirteen polynomials has degree 2976. Their difference is 744, yielding the additional polynomial $\Phi_{744}(x)$, whose degree is 240.

These identities form a precise mathematical construction. They may serve as a starting point for further research into formal representations, algorithms, statistical systems, and institutional coordination. They do not independently determine economic gains, legal authority, political legitimacy, or geopolitical outcomes.

A credible framework of regional cooperation must combine mathematical clarity with empirical discipline, reciprocal benefit, sovereign equality, lawful conduct, and voluntary participation. Its institutional merits must ultimately be judged by evidence and by the welfare and rights of the people affected.

A Summary of Principal Results

Table 2: Summary of principal results.

Quantity	Value
Number of represented countries	13
Sum of selected cyclotomic indices	3720
Sum of adjusted polynomial degrees	2976
Degree of aggregate product $\mathcal{P}(x)$	2976
Difference between index sum and product degree	744
Prime factorisation of 744	$2^3 \cdot 3 \cdot 31$
Degree of $\Phi_{744}(x)$	240

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Glossary

Cyclotomic polynomial
: The monic polynomial whose roots are the primitive roots of unity of a specified order.

Euler's totient function
: The function $\varphi(n)$, counting the positive integers up to n that are relatively prime to n.

Polynomial degree
: The highest exponent with a nonzero coefficient in a nonzero polynomial.

Index
: In $\Phi_n(x)$, the positive integer n specifying the order of the primitive roots of unity.

Aggregate polynomial
: The product $\mathcal{P}(x)$ of the thirteen selected cyclotomic polynomials.

Statistical harmonisation
: The alignment of definitions, measurement units, and reporting procedures so that data from different sources can be compared meaningfully.

Econometric identification
: The assumptions and design conditions required to infer a causal or structural relationship from observed data.

Welfare economics
: The study of how economic arrangements affect welfare, distribution, and the evaluation of social outcomes.

Reciprocity
: A principle under which cooperation and obligations are designed to recognise the interests and contributions of participating parties.

Sovereign equality
: The principle that states possess equal legal status under international law, subject to the applicable international legal framework.

Model validation
: The process of testing whether a model's calculations, assumptions, and predictions are reliable for its intended purpose.

The End